Ritsumeikan Asia Pacific University

College of Sustainability and Tourism

Graduation Thesis

Three Margins of Electricity Recovery

Coverage, Reported Entry, and Generator Sizing in Japan's Municipal Incinerator Fleet, FY2005–FY2024

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Abstract

Facility counts can misdiagnose electricity recovery when they collapse fleet coverage, entry, and engineering performance into one statistic. This thesis deterministically links 23,593 Japanese municipal-incinerator records from fiscal year (FY) 2005 to FY2024 into 1,690 stable administrative lineages and 1,767 reported asset episodes. In FY2024, 41.1% of all records reported installed electrical capacity, but positive-output facilities handled 80.1% of recorded waste throughput. Facilities reporting installed capacity represented 70.5% of processing design capacity. The all-record installed-capacity share rose by 19.5 percentage points from FY2005, whereas the rise was 2.2 points among 732 lineages observed at both endpoints. This contrast is consistent with a substantial contribution from changing fleet composition. First reported capacity entry was rare: the primary discrete-time model contained 35 events. At support-rich capacities of 24, 60, and 120 tonnes per day, standardized annual risks were 0.68, 1.37, and 3.29 entries per 1,000 facility-years. A five-parameter Firth model gave a 300-versus-100 tonnes-per-day odds ratio of 6.72 (1,999-lineage-bootstrap 95% confidence interval: 4.31–12.46), although 300 tonnes per day was a thinly supported tail value. The positive scale association survived continuity, functional-form, reporting-state, leave-one-prefecture influence, and event-influence checks. Among 6,511 engineering-valid generator-years from 493 lineages, adjusted installed capacity was 79.1%, 58.6%, and 23.5% lower in pre-1990, 1990s, and 2000s start-year cohorts than in the 2010-or-later cohort. Administrative annual capacity-factor proxies were not uniformly lower in older cohorts. Under common controls, installed sizing was the largest absolute point-estimate accounting component of each older cohort's log gross-generation-intensity gap. The exact component sum is an identity, not independent causal evidence. The contribution is a three-margin national diagnosis; it does not identify physical projects, recoverable potential, or causal effects. Independent clerical review of the highest-risk lineage links remains a human validation gate.

Keywords: municipal solid waste; incineration; waste-to-energy; Japan; generator sizing; capacity factor; administrative record linkage

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1 Introduction

Japan's municipal waste system relies extensively on incineration for volume reduction and hygienic treatment. Electricity recovery can use part of the heat released during combustion, but it does not by itself make incineration preferable to prevention, reuse, or recycling. Its value depends on plant design, waste properties, energy use, and the wider waste and electricity systems (Astrup et al., 2009, 2015; Brunner & Rechberger, 2015; European Commission, 2017). Japan-specific work likewise extends beyond a binary label of whether a plant generates electricity (Uno, 2015; Tabata & Tsai, 2016; Yamada et al., 2023).

The Ministry's FY2024 national summary reports 415 electricity-generating facilities among 991 incineration facilities, or 41.9% (Ministry of the Environment Japan, 2026). That published context is distinct from this thesis's analytical measure: 417 of 1,014 retained FY2024 facility records report positive installed electrical capacity, or 41.1%. The numerator definition and record denominator differ, so the official ratio is not substituted into the analytical calculations.

The most visible national statistic is often a facility count. Yet a small intermittent facility and a large continuous facility each contribute one unit, despite handling different waste volumes. A count describes how widely equipment appears across records, not how much waste passes through generating facilities or where processing capacity is located. Those are distinct coverage questions.

A second problem arises in facility histories. First reporting of installed capacity is uncommon and can occur within an administratively continuous facility, with a reported asset reset, or in a forward-dated record. These cases do not have one verified physical meaning. The workbooks also lack a reliable identifier over the full period: codes are absent in FY2010–FY2012 and fully change between FY2019 and FY2020. Treating them as persistent keys would break histories and create false events.

A third problem concerns gross electricity divided by throughput. This gross megawatt-hours-per-tonne (MWh/t) ratio combines generator size relative to processing capacity, annual use of installed electrical capacity, and waste loading. Treating it as an independent operating measure can make a design-cohort difference look operational.

This thesis addresses the three problems in one national facility-level study. It reconstructs stable administrative lineages before creating lags, separates facility participation from waste-volume and design-capacity coverage, uses a bias-reduced event-history model for sparse first-entry events, and decomposes gross generation intensity into observable engineering components. The aim is descriptive and diagnostic. The study does not estimate the effect of a specific equipment project, infer a physical closure from administrative absence, or calculate net electricity export, useful heat, lifecycle emissions, or recoverable technical potential.

1.1 Research questions

The three research questions (RQs) are:

> **RQ1: Coverage and fleet composition.** How did the share of facility records reporting installed electrical-generation capacity evolve from FY2005 to FY2024; how does that count-based measure compare with the shares of recorded waste throughput and processing design capacity covered by generating facilities; and how much of the endpoint change remains among administrative lineages observed in both years?
>
> **RQ2: First reported installed-capacity entry.** Among stable administrative lineages first observed without installed electrical-generation capacity, which prior-year age and waste-processing-scale profiles are associated with first reporting positive capacity, and does the evidence change after the risk set is restricted to lineages with positive prior-year waste throughput?
>
> **RQ3: Installed design and annual use.** Among engineering-valid operating-generator years, how do raw installed electrical capacity and annual electrical capacity factor differ across reported facility start-year cohorts after observed controls? Under a common control design, what are the relative accounting components of cohort differences in gross MWh/t?

The sequence matters. RQ1 establishes the denominator problem at fleet level. RQ2 then asks which observed non-generating lineages enter the installed-capacity state. RQ3 begins only after positive throughput and gross output are observed, and asks what produces differences within the generating segment. Figure 1 shows the annual count, throughput, and design-capacity coverage measures that anchor this sequence.

Reader map for the three research questions

RQ	Analysis population	Evidence approach	Interpretation boundary
RQ1	Full retained annual facility panel	Compare facility participation, throughput coverage, design-capacity coverage, and endpoint-common change.	Does not measure conversion efficiency, environmental optimality, or causal retrofit.
RQ2	Administrative lineages at risk of first reported installed-capacity entry	Use a sparse-event Firth model to profile entry by prior-year age and processing scale.	Does not identify a physical project date or the causal effect of changing age or scale.
RQ3	Engineering-valid operating-generator years	Model raw installed kW and annual capacity factor, then reconcile the components of gross MWh/t.	Does not measure net export, thermodynamic or lifecycle efficiency, or causal mediation.

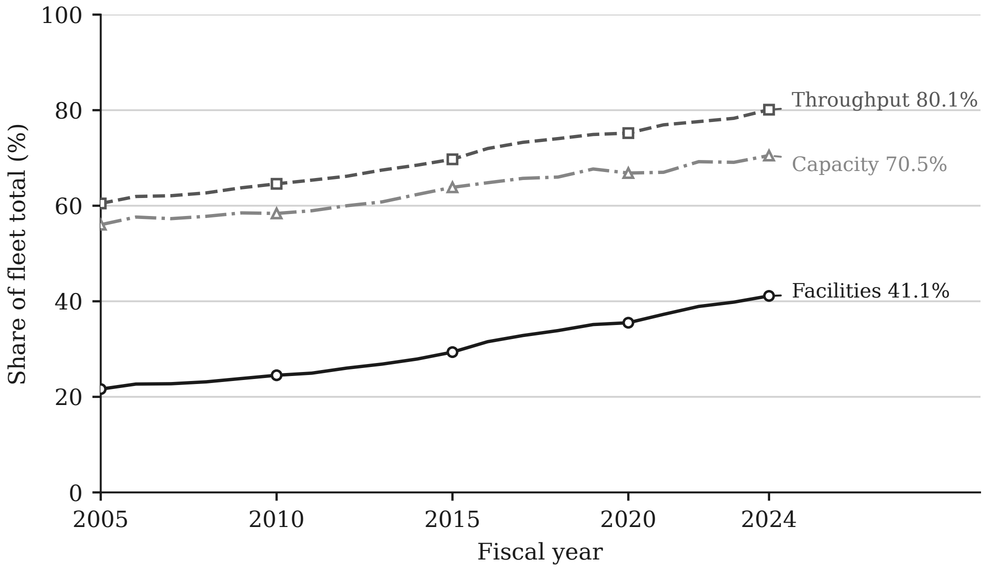

Figure 1: Annual electricity-recovery coverage among retained Japanese municipal-incinerator records, FY2005–FY2024. "Facilities" is the share of records reporting positive installed electrical capacity; "Throughput" is the share of annual waste processed at records reporting positive electricity output; and "Capacity" is the share of total waste-processing design capacity at records reporting positive installed electrical capacity. These three denominators answer different questions and are not interchangeable.

1.2 Thesis objectives and significance

The thesis has four linked objectives. First, it establishes a reproducible longitudinal facility record before calculating transitions. This is necessary because the official identifiers do not provide an uninterrupted key over the study period. Second, it compares three fleet-coverage denominators rather than allowing the facility count to stand for the whole system. Third, it estimates first reported installed-capacity entry only within an explicit population at risk and uses a sparse-event estimator suited to the small number of observed entries. Fourth, it decomposes gross generation per tonne into generator design intensity, electrical capacity factor, and waste-processing utilization. The objectives therefore progress from data identity, to fleet description, to transition, and finally to conditional engineering structure.

The academic significance lies primarily in this integrated measurement architecture. Existing studies offer important analyses of system outcomes, facility performance, and Japanese waste-to-energy policy, but those questions do not share one denominator or one comparison population. Treating them as if they did can turn a descriptive fleet share into a performance statement, or a conditional generator comparison into an explanation of adoption. This thesis shows how the three questions can be studied together without collapsing their estimands. It also demonstrates why identity reconstruction is part of the research design rather than a preliminary clerical step: transition estimates depend on which annual records are considered continuous.

The practical significance is diagnostic rather than prescriptive. National and municipal decision-makers need to know whether low participation by count also means low coverage of waste activity, whether observed entry is broadly distributed or concentrated among particular facility profiles, and whether a gross-output difference reflects generator size or annual use. The results can identify where additional engineering, financial, or project-history evidence is most valuable. They cannot by themselves select a retrofit, replacement, or closure project.

The social significance follows from keeping electricity recovery within the waste hierarchy. Recovering electricity can reduce wasted combustion heat, but it does not make waste prevention, reuse, and recycling unnecessary. Clear measurement helps prevent a high throughput-coverage statistic from being presented as proof of environmental optimality. It also avoids treating every non-generating facility as an equally feasible opportunity when local waste supply, costs, grid access, heat demand, and alternative treatment arrangements remain unobserved.

2 Literature Review and Analytical Foundation

2.1 Waste-to-energy within the waste hierarchy

Waste-to-energy (WtE) is not a single environmental outcome. It is a family of thermal-treatment arrangements that can provide waste-disposal service, recover electricity or useful heat, and produce emissions and residual materials at the same time. Reviews therefore evaluate incineration within material-flow, life-cycle, energy-system, and waste-hierarchy boundaries rather than by electricity output alone (Astrup et al., 2009, 2015; Brunner & Rechberger, 2015; Lombardi et al., 2015; Münster & Meibom, 2010). The environmental value assigned to recovered energy can change with waste composition, the energy displaced, plant configuration, treatment alternatives, and the boundary chosen for analysis.

This wider framing is especially important for a study of an incineration-heavy system. Comparative policy research places prevention, reuse, and material recycling ahead of disposal, while recognizing energy recovery as one possible function of residual-waste treatment (European Commission, 2017; Sakai et al., 2011). Increasing electricity recovery is therefore not equivalent to improving the whole waste system. More generation could reflect better use of unavoidable combustion heat, but it could also coexist with high residual-waste throughput. The appropriate interpretation depends on which question is being asked.

This thesis deliberately conditions on the observed Japanese incineration fleet. It asks how electricity recovery is distributed within that fleet, not whether incineration should replace prevention or recycling. It also does not estimate life-cycle greenhouse-gas savings, avoided generation, pollutant exposure, ash management, heat use, or social welfare. Keeping these outcomes outside the estimand prevents the gross electricity measures reported below from becoming an unsupported claim of overall sustainability.

2.2 Japan's municipal transition and operating context

Japan's municipal solid-waste system combines national policy with local implementation. Municipal instruments can alter the quantity and composition of residual waste before it reaches an incinerator. For example, Japanese unit-charging programmes have been studied as waste-reduction and recycling measures, with effects depending on programme design and complementary collection arrangements (Sakai et al., 2008). This means that annual incinerator throughput is not simply an engineering input: it is partly the downstream result of local waste-management choices that are not fully observed in the facility workbooks.

Within the treatment stage, prior Japan-focused research examines several different margins. Uno (2015) describes technical developments associated with higher-efficiency power generation. Tabata and Tsai (2016) examine heat supply and the local constraints on using recovered heat. Sun et al. (2018) show through a Tokyo case that energy recovery can be considered within an integrated urban waste-management network rather than at one plant in isolation. Sasao (2018) uses repeated facility observations to study how municipal solid-waste policy relates to heat and electricity output. Shino (2019) examines generation performance relative to waste input, while Yamada et al. (2023) place the sector within longer-run Japanese decarbonization scenarios.

The institutional setting also changed during the observation window. The Ministry maintains a high-efficiency waste-power facility development manual that links facility-support requirements to generation performance and plant planning (Ministry of the Environment Japan, n.d.). The Great East Japan Earthquake and Fukushima Daiichi accident in March 2011 then changed the national energy-policy context. Japan's feed-in tariff (FIT) system began in July 2012, and waste-derived biomass electricity was among the eligible renewable categories (Agency for Natural Resources and Energy, 2017; Sasao, 2018). In FY2014, the Ministry also offered capital support for high-efficiency waste-heat recovery and waste-derived biomass generation (Ministry of the Environment Japan, 2014).

These milestones make calendar time substantively relevant, but do not create a causal design here. The workbooks do not identify subsidy receipt, FIT accreditation, electricity contracts, or investment motives. The entry model therefore includes a smooth calendar term; it does not treat FY2011 or FY2012 as an exogenous breakpoint or estimate a Fukushima or FIT effect.

Scale is important in this literature, but it is not a complete decision rule. Large, continuously operated plants can support steam conditions and generating equipment that are difficult to reproduce at small sites. Yet a heat-balance case study by Yoshida et al. (2018) identifies technically possible recovery options for a small Japanese facility and separately examines their costs and benefits. The relevant question is therefore not simply whether a small plant lacks a generator. Feasibility can also depend on waste supply, technology, capital and operating costs, grid connection, heat demand, and regional coordination.

The national administrative panel used here does not consistently observe those project-level factors. It can show which reported facility profiles are associated with witnessed entry, but it cannot identify why a municipality made a particular investment. This boundary is consequential: an association between size and entry is evidence about observed selection in the fleet, not proof that size alone caused adoption or that every large non-generator should be retrofitted.

2.3 Measuring coverage, performance, and efficiency

The literature also shows why the word *efficiency* requires qualification. Thermal-treatment reviews compare technologies using electrical, thermal, or combined energy performance and sometimes life-cycle measures (Astrup et al., 2015; Lombardi et al., 2015). The European Union's R1 energy-recovery formula uses a defined accounting boundary (Grosso et al., 2010). Facility benchmarking studies may instead estimate technical or revenue efficiency relative to observed peers (Chen et al., 2012; Yeh, 2020). These are not interchangeable quantities.

The Ministry workbooks report installed electrical capacity, annual gross electricity generation, waste-processing design capacity, and annual throughput. They do not provide a complete and consistent series for net export, internal electricity use, useful heat delivery, lower heating value, steam conditions, operating cost, or emissions. Gross megawatt-hours per tonne (MWh/t) is therefore observable, but it is not a direct thermal-conversion efficiency. Shino (2019) similarly treats generation relative to waste input as informative while noting that a thermal interpretation requires calorific-value information.

Measurement also changes the apparent reach of energy recovery. A facility-count share weights a small and a large plant equally. A throughput share weights each plant by the tonnes it actually processes. A design-capacity share weights its nominal daily waste-processing capacity. None is inherently the correct denominator for every purpose. Used together, however, they reveal whether generation is diffuse across facilities or concentrated where most waste is processed and capacity is installed.

The present design consequently separates three estimands. Fleet coverage is an annual ratio whose denominator is facilities, tonnes, or waste-processing design capacity. Entry is a discrete-time conditional probability among lineages observed at risk. Generator-component estimates are conditional associations among positive-throughput, positive-output generator-years that pass stated engineering checks. A coefficient from the generator frame cannot explain why a non-generator enters, and a facility participation rate cannot describe the share of waste processed with electricity output.

This separation permits a low count share and high throughput share to be true simultaneously. It also permits newer cohorts to report higher gross MWh/t because of larger generators even when older generators do not have lower annual capacity factors. Section 3 states the exact identity used to distinguish those components.

2.4 Infrastructure heterogeneity, transition, and persistence

Incinerators are long-lived infrastructure embedded in municipal organisations, regulation, collection systems, energy networks, and local service obligations. Socio-technical research distinguishes a technology from the wider system of actors, rules, users, and material arrangements that supports it (Geels, 2004). Carbon lock-in research likewise explains how long-lived capital and institutional interdependence can make infrastructure pathways persistent (Seto et al., 2016; Unruh, 2000). These perspectives make age, cohort, and continuity relevant, but they do not establish that every old incinerator is technically or politically locked in.

This thesis uses that literature as a conceptual warning rather than a tested causal theory. The workbooks do not observe investment deliberations, sunk costs, procurement contracts, or political opposition. They also do not show whether first reported generation was installed through an in-place retrofit, a furnace replacement, a wider site redevelopment, or correction of a previous record. Accordingly, the longitudinal unit is called a stable administrative lineage, and large configuration changes create separate asset episodes. A first-entry event means the first witnessed transition from no reported generation to reported generation under a stated continuity rule; it is not automatically labelled a retrofit.

This distinction explains why identity reconstruction is substantive. If a national recoding is mistaken for facility replacement, apparent entry and exit events will be manufactured by the database. If all records sharing a site name are forced into one permanent asset, genuine replacement may be hidden. The lineage and episode sensitivity analyses translate the general infrastructure literature into a testable question: how much do transition inferences depend on the continuity definition?

2.5 Methodological comparators and adaptation

Recent studies supply research logic, not a ready-made model. Cui et al. (2026) foreground facility hierarchy, Liu et al. (2025) emphasize effectiveness over expansion, and Han et al. (2025) place recovery beside other sustainability dimensions. This thesis asks where hierarchy appears in Japan without estimating an optimization frontier, city energy-carbon system, or pollutant outcomes.

Sasao (2018) supports repeated Japanese facility observations and explicit output questions. Shino (2019) makes generation relative to waste input an important observable. This thesis retains that observable but treats it as an accounting intensity requiring decomposition.

Chen et al. (2012) and Yeh (2020) motivate separating activities within operating plants. This study adds an entry risk set and uses component regressions because the available fields do not support a comparable network or revenue frontier across all years.

Table 1: Comparator gap matrix and transparent method adaptation

Study	Unit and time structure	Primary outcome	Identity treatment	Gap retained here
Cui et al. (2026)	975 Chinese plants and 2,151 incinerators; 2023 operations and scenarios to 2035	Efficiency hierarchy and optimization	Plant/line database; entry histories not targeted	Locate hierarchy in Japan without importing an optimization frontier
Liu et al. (2025)	Chinese urban waste-energy-carbon systems	Effectiveness versus expansion	City systems, not facility linkage	Distinguish equipment diffusion from covered activity
Han et al. (2025)	Chinese configurations and recovery outcomes	Pollutant control and recovery	Configuration comparison	Keep electricity separate from unobserved pollutant and lifecycle outcomes
Sasao (2018)	Repeated Japanese incinerator observations	Policy, technology, heat, and electricity	Repeated facilities; first entry not targeted	Reconstruct lineages and an explicit non-generator risk set
Shino (2019)	22 Tokyo incinerators, FY2012–FY2017	Generation per waste input and combustion conditions	Operating-plant comparison	Decompose gross MWh/t rather than use one score
Chen et al. (2012); Yeh (2020)	Operating incineration plants	Multi-activity technical or revenue performance	Persistent plants within performance samples	Use observable components without consistent frontier inputs

The adaptation changes the unit and estimands to fit Japanese administrative data. Cui et al. supply the hierarchy logic, Sasao the closest Japan-wide repeated-facility precedent, Shino the transparent gross-generation-per-input measure, and component-performance studies the reason to separate activities. This thesis does not copy their outcome models. It reconstructs the longitudinal comparison population first and asks questions those designs leave unresolved.

2.6 Synthesis, research gap, and contribution

The reviewed literature provides strong but separated accounts of the problem. Waste-system studies define environmental and hierarchy boundaries. Japan studies document policy, technology, heat use, integrated recovery, and facility-level output. Performance studies distinguish engineering indicators from peer-relative efficiency. Transition research explains why infrastructure continuity and institutional context may matter. Close empirical comparators show the value of repeated facility observations and decomposition. Each strand answers a necessary part of the problem, but their units and outcomes differ.

The novelty claim is substantive before procedural. First, a facility count understates how much observed activity is covered. The much smaller increase among endpoint-common lineages is consistent with a substantial contribution from changing observed fleet composition; it is not a formal additive decomposition. Second, first reported entry is scale-associated after sparse- event, continuity, functional-form, reporting-state, geographic, and influence checks. Third, the apparent start-cohort hierarchy in gross MWh/t is consistent with a substantial installed-sizing contribution rather than uniformly better annual use. These reinterpretations change what familiar national statistics mean.

Within the literature reviewed here, the bounded integration claim is that these reinterpretations have not been connected for Japan over FY2005–FY2024 using deterministically reconstructed lineages across coding breaks, denominator-matched coverage, an explicit first-entry risk set, and an exact engineering identity. This is not a systematic-review claim and does not assert that the individual methods are new. The contribution is the resulting diagnosis under one traceable set of estimands.

What this design changes. Without the three-margin separation, the same administrative records can support three tempting but incomplete conclusions: that participation below one half means little waste is covered, that rising participation mainly represents conversion among incumbent facilities, and that higher gross MWh/t means uniformly better annual operation. The thesis tests each interpretation against the denominator, continuity rule, or engineering component it omits. Coverage, transition, and conditional generator structure therefore operate as complementary diagnostics, not substitutes for one another and not a causal policy evaluation.

3 Data and Methods

3.1 Source workbooks and recoverable provenance

The source is the Ministry of the Environment Japan General Waste Treatment Survey facility workbook series, also distributed through the Japanese official statistics system (e-Stat, n.d.; Ministry of the Environment Japan, 2026). The study covers 20 fiscal years, FY2005–FY2024. For each year, the checked-in parser reads the first workbook sheet and searches the first six spreadsheet rows for recognized Japanese header terms. The resulting standardized fields include facility and municipality information, reported start year, waste-processing design capacity, annual throughput, furnace and facility attributes, installed electrical capacity, and gross electricity generated.

The provenance audit records all 20 workbooks, their filenames, byte sizes, 256-bit Secure Hash Algorithm (SHA-256) hashes, parsed sheet names, detected header rows, and final field mappings. The files total 15,158,836 bytes and have 20 distinct hashes. Seventeen standardized fields are detected in the older workbooks; 19 are detected from FY2018 onward. Sold-electricity and sales-revenue fields are not detected for FY2005–FY2017, which is one reason the outcome is gross generation rather than net export or revenue. The original download timestamp was not recorded. Volatile checkout modification times are deliberately not persisted; the manifest instead records each workbook's last Git commit timestamp as repository history, not retrieval time. Configured source URLs were not revalidated during the provenance build. The hashes establish which local files produced the results, not the publisher's revision history or a complete acquisition chain.

Parsing yields 23,599 raw rows. Six rows are exact duplicates of another source record and are collapsed before identity resolution, leaving 23,593 unique facility-year records. Derived age is fiscal year minus reported start year. Thirty-six source ages are missing and 355 are negative. Negative values are converted to missing for age-dependent analysis rather than set to zero. Waste-processing utilization is annual throughput divided by 365 times design capacity. Installed generation is indicated by reported electrical capacity greater than zero.

3.2 Stable administrative lineages and asset episodes

Official facility codes cannot support the longitudinal design by themselves. All 3,716 rows in FY2010–FY2012 lack a code. Between FY2019 and FY2020, both years contain codes, yet the sets have zero overlap. The latter break is a complete recode, not evidence that the entire fleet was replaced. Across that transition, the identity resolver restores 1,064 adjacent-year lineage links, equivalent to 97.3% of FY2019 records. Across the longer FY2009–FY2013 bridge, 882 official codes overlap while 1,135 stable lineages are linked.

The resolver therefore constructs a *stable administrative lineage*: a sequence of records judged to describe the same continuing administrative facility or site history. The implementation field is `stable_site_id`, but the term does not assert that ownership, buildings, furnaces, or other physical assets remain unchanged. The algorithm is deterministic and proceeds as follows.

1. Text and identifiers are normalized conservatively, including Unicode form, spacing, punctuation, and numeric-code formatting. A complete standardized row fingerprint identifies exact duplicate records, which are collapsed before matching.

2. Records are compared only within prefecture. Candidate evidence combines normalized facility name, municipality, reported start year, waste-processing capacity, furnace count, facility type, and the annual official code. Code agreement is supporting evidence, but contradictory name and configuration evidence can veto a code match.

3. Adjacent fiscal years are resolved before gaps, preventing an older history from winning a tie over an otherwise equivalent immediately prior record. Short gaps of up to four years are considered afterward. Within each prefecture-year problem, one-to-one global assignment prevents one prior record from being assigned to multiple current records.

4. Unmatched records seed new deterministic lineage IDs from canonical record fingerprints rather than row order. Within a lineage, a new asset episode is created when reported start year changes materially in either direction, a mature record resets to a near-zero age, or a major name and configuration discontinuity appears.

The result contains 1,690 stable administrative lineages and 1,767 asset episodes, with no duplicate lineage-year and no history longer than the 20-year study window. The match audit classifies 15,324 records as code plus exact-name links, 274 as code with other supporting evidence, 6,204 as exact-name links without code support, 101 as fuzzy-name links without code support, and 1,690 as new lineage seeds.

Candidate links below the minimum score are excluded before assignment, as are ambiguous links lacking an exact-name or official-code signal. This removes 3,092 subthreshold and 15,308 weak ambiguous candidate edges. Sixteen accepted links, spanning 14 lineages, still have a low current-record or prior-record margin but retain strong evidence; every one is exposed in the identity audit. The entry and component analyses therefore include whole-lineage identity-certain sensitivities that exclude all 14 affected lineages rather than selectively removing individual years.

Known difficult cases are embedded as executable checks. Three pairs that must link and three pairs that must remain separate are tested directly, including examples from the FY2019–FY2020 recode and earlier code reuse. The resolver is also rerun after random row permutation and after insertion of an unrelated synthetic record in six difficult or duplicate-bearing prefectures. The lineage/episode mapping must remain unchanged. These golden-link, separation, permutation, and insertion tests reduce identifiable implementation risks; they do not make probabilistic record linkage certain. Any result depending on a small number of lineages remains vulnerable to unresolved name or configuration ambiguity. The uncertainty flag makes that vulnerability testable rather than implicit.

A blinded clerical-review packet contains all 35 modeled event links, all 16 accepted uncertain links, all 31 gap links, 50 FY2019–FY2020 bridges, and a stratified accepted-pair sample (Harron et al., 2020). Scores, decisions, and final lineage IDs are withheld. Until a second reviewer completes it and disagreements are adjudicated, the packet is not independent validation. Completion and adjudication remain the principal outstanding linkage-validation step before the reconstructed lineages are described as independently reviewed.

3.3 Analytical frames and estimands

The three RQs use related but non-identical samples. Table 2 prevents their denominators from being blended.

The entry sample excludes 467 lineages already reporting positive installed capacity

Table 2: Analytical frames after identity and data-quality checks

Frame	Facility-year rows	Stable lineages	Events	Estimand
Full administrative panel	23,593	1,690	–	Annual facility, throughput, and design-capacity coverage
Descriptive non-generator risk set	16,519	1,223	55	Observed first installed-capacity entries during follow-up
Broad exact-year complete-covariate entry model	15,154	1,137	35	Conditional annual administrative-lineage entry odds using prior-year covariates
Exact-year model after prior operation	13,072	1,019	33	Nested sensitivity requiring positive prior-year throughput
Same-asset-episode continuity sensitivity	15,095	1,135	24	Entry odds after excluding transitions that cross an inferred asset-episode boundary
Identity-certain-lineage sensitivity	15,107	1,130	35	Broad model after excluding every lineage containing an accepted uncertain identity link
Two-prior-year reporting-state sensitivity	14,000	1,110	30	Requires two observed prior years with neither positive reported capacity nor output
Positive-throughput, positive-output generators	6,660	504	–	Descriptive operating-generator frame
Engineering-valid component model	6,511	493	–	Conditional generator-component associations

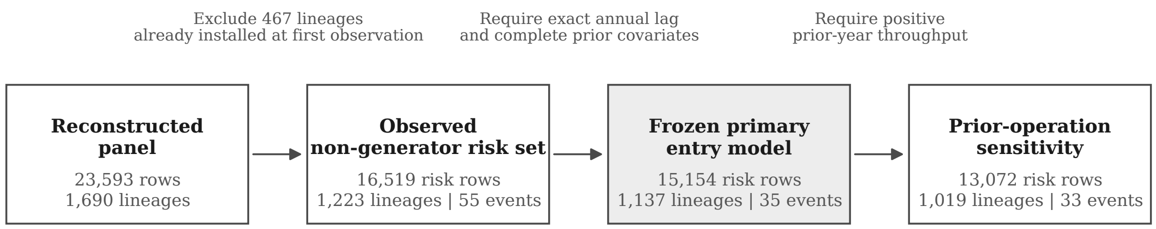

Figure 2: Entry-model sample construction. The reconstructed panel contains 23,593 facility-years from 1,690 administrative lineages. Restricting attention to lineages first observed without positive installed capacity produces 16,519 risk rows and 55 observed entries; exact annual lags and complete prior covariates define the 15,154-row, 35-event primary model. The prior-operation sensitivity further requires positive prior-year throughput. The 467 lineages already installed at first observation are left-censored, not non-adopters.

in their first observed year because their entry time is left-censored. A remaining lineage contributes annual risk rows until its first positive capacity observation or its last observed non-generating year. The first observed risk row is dropped from modeling because a lagged predictor is not available. The complete-covariate frame further drops 120 rows with missing lagged age or processing capacity, and 22 non-adjacent rows are excluded from the exact-year model; none of those 22 rows contains an event.

The 55 descriptive events, the 35 complete-covariate events, and the pathway categories answer different bookkeeping questions. Coincidentally, the pathway audit also identifies 35 continuity-lineage events. That count should not be conflated with the 35 events in the exact-year regression sample. The former is a pathway classification among observed entries; the latter is a covariate completeness restriction.

3.4 RQ1: coverage definitions and fleet composition

For fiscal year t, let N_t be all retained facility records, I_{it}^K indicate positive installed electrical capacity, I_{it}^G indicate positive gross generation with positive throughput, I_{it}^W indicate positive throughput, W_{it} be annual throughput, and C_{it} be waste-processing design capacity. To isolate denominator effects from state- definition effects, facility participation is reported as a matched two-by-two matrix:

$$P_t^{K,\text{all}} = \frac{\sum_i I_{it}^K}{N_t}, \qquad P_t^{K,\text{active}} = \frac{\sum_i I_{it}^K I_{it}^W}{\sum_i I_{it}^W}, \tag{1}$$

$$P_t^{G,\text{all}} = \frac{\sum_i I_{it}^G}{N_t}, \qquad P_t^{G,\text{active}} = \frac{\sum_i I_{it}^G}{\sum_i I_{it}^W}. \tag{2}$$

Holding the numerator state fixed makes the all-record versus active-record contrast a denominator comparison. Holding the denominator fixed shows the small difference between reported installed capacity and reported positive output. The activity-weighted coverage measures are

$$P_t^{\text{throughput}} = \frac{\sum_i W_{it} I_{it}^G}{\sum_i W_{it}}, \tag{3}$$

and

$$P_t^{\text{design}} = \frac{\sum_i C_{it} I_{it}^K}{\sum_i C_{it}}. \tag{4}$$

The throughput measure uses tonnes and the design measure tonnes per day. Positive output is used in the throughput numerator because installed capacity does not guarantee positive annual generation. For the engineering-valid subset, fleet gross intensity has the exact decomposition

$$\frac{\sum_i G_{it}^{\text{valid}}}{\sum_i W_{it}} = \left(\frac{\sum_i W_{it}^{\text{valid}}}{\sum_i W_{it}}\right)\left(\frac{\sum_i G_{it}^{\text{valid}}}{\sum_i W_{it}^{\text{valid}}}\right). \tag{5}$$

The first term is valid generator-throughput coverage and the second is throughput-weighted conditional gross intensity. This identity distinguishes how much waste is covered from how much gross electricity is reported per tonne within the covered segment.

The annual percentages are repeated cross-sections, not a direct measure of conversion among incumbent facilities. Let S_t be the stable administrative lineages observed in year t and $S^{\text{common}} = S_{2005} \cap S_{2024}$. Endpoint-common prevalence is

$$P_t^{K,\text{common}} = \frac{\sum_{i \in S^{\text{common}}} I_{it}^K}{|S^{\text{common}}|}, \qquad t \in \{2005, 2024\}. \tag{6}$$

Two stricter diagnostics retain lineages with the same reported asset episode at both endpoints or lineages observed in all 20 fiscal years. Endpoint-only groups show how different the observed compositions are, but administrative appearance and disappearance are not called physical openings or closures. Because these groups use different denominators, the comparison is not an additive causal decomposition. It asks whether the all-record trend resembles widespread within-lineage state change or changing fleet-record composition.

3.5 RQ2: sparse first-entry model

3.5.1 Event and population at risk

In plain terms, each eligible non-generator lineage contributes one row for each year in which it remains observed and has not previously reported positive installed capacity. The outcome records whether that lineage first reports positive capacity in the current year. This structure compares entry profiles among lineages that could still experience a first observed entry; it does not compare all facilities indiscriminately.

The event is the first observed year with positive installed electrical capacity after a lineage has been observed without it. It is an administrative state transition. It does not uniquely identify first physical operation, equipment installation, or a particular construction history.

Positive installed capacity means a non-missing reported value greater than zero. A blank or zero field means no reported positive capacity; it is not verified physical absence. In total, 49 panel rows across 6 lineages report positive gross output with blank or zero installed capacity. None of the 35 modeled events reports positive output in its immediately prior year, and 32 report positive output in the event year. A stricter sensitivity requires two observed prior years with neither positive reported capacity nor output; it retains 14,000 rows, 1,110 lineages, and 30 events.

3.5.2 Primary model, estimator, and uncertainty

The model asks a narrow question: among lineages still at risk, how do the odds of first reported entry vary with prior-year age and processing scale after accounting for calendar time and elapsed observed time at risk? It describes those profiles; it does not estimate what would happen if a municipality physically enlarged or aged a facility.

Discrete-time event-history analysis represents each at-risk lineage-year as a binary outcome (Allison, 1982; Beck et al., 1998). For lineage i in year t, let $H_{it} = 1$ indicate that the lineage remains in the first-entry risk set at the start of the year because no earlier observed year reported positive installed capacity. The revision-frozen primary model is

$$\operatorname{logit}\{\Pr(Y_{it}=1 \mid H_{it}=1)\} = \alpha + \beta_A \frac{A_{i,t-1}}{10} + \beta_C \log\left(1+\frac{C_{i,t-1}}{100}\right) + \beta_T \frac{t-2014.5}{5} + \beta_R \log(1+R_{it}). \tag{7}$$

Here, $Y_{it}=1$ denotes first reported positive installed capacity, A is prior-year reported age, C is prior-year processing design capacity, and R is elapsed observed time at risk. Processing capacity is measured in tonnes per day (t/day). Age is scaled per ten years and calendar time per five years. Including the intercept, the model has five parameters for 35 broad-frame events. An internal model-decision memo documented this specification before revised estimates were compared. It was not externally preregistered or prospectively registered; the memo establishes the internal revision sequence, not confirmatory status. The earlier 11-parameter model is retained as sensitivity. Technology and geography are not added because 35 events cannot support a larger adjustment set reliably. Their omission is a limitation: processing scale can absorb unobserved technology, regional, financial, or municipal differences. Leave-one-prefecture fits assess concentration and influence only; they are not geographic- confounding control.

Ordinary maximum-likelihood logit is vulnerable to small-sample bias and separation in a sparse event design. The primary estimator therefore uses Firth's bias reduction (Firth, 1993) and its finite-estimate solution to separation (Heinze & Schemper, 2002), maximizing the penalized log-likelihood

$$\ell_F(\theta) = \ell(\theta) + \frac{1}{2}\log\left|\mathcal{I}(\theta)\right|, \tag{8}$$

where ℓ is the binomial log-likelihood and $\mathcal{I}$ is the expected information matrix. The Jeffreys-prior penalty reduces first-order bias and keeps finite estimates under separation. The coefficient table labels the fitted-model standard errors, confidence intervals, and term-level p-values as model-based. To represent repeated observations from the same lineage, the primary uncertainty calculation uses 1,999 deterministic cluster-bootstrap replications that resample complete lineages with replacement. Percentile intervals use the resulting coefficient distributions. Every requested replication must converge and return all focal coefficients.

Four frames are fitted. The broad exact-year frame is primary and includes every complete at-risk row whose lag belongs to the same stable administrative lineage; it can cross an inferred asset-episode boundary because its estimand is first administrative-lineage entry. The prior-operation frame is a nested sensitivity that additionally requires positive prior-year throughput. It is not an independent comparison group, so differences between it and the broad frame are not treated as an equality or equivalence test. A same-asset-episode continuity sensitivity excludes the 59 exact-year rows that cross an inferred episode boundary, including 11 events. An identity-certain sensitivity excludes every lineage containing any of the 16 accepted uncertain links. The prior-operation, same-asset-episode, and identity-certain frames are sensitivity analyses rather than co-primary models.

3.5.3 Translating the scale association

Odds ratios show relative differences but can obscure how rare entry remains. The thesis therefore reports both the internally frozen 300-versus-100 t/day odds ratio and standardized annual risks at scale values with substantially greater empirical support.

For an intuitive scale contrast, the odds ratio comparing 300 with 100 t/day is

$$\begin{aligned} OR_{300:100} &= \exp[\beta_C \{\log(1+300/100) - \log(1+100/100)\}] \\ &= \exp(\beta_C \log 2). \end{aligned} \tag{9}$$

Because an odds ratio does not show how uncommon entry remains, standardized annual probabilities are also reported. For capacity level c, the fitted probability is averaged over the observed broad-frame distribution of age, calendar year, and elapsed risk:

$$\bar{p}(c) = \frac{1}{n} \sum_{i,t} \operatorname{logit}^{-1}\left\{\mathbf{x}_{it}(c)'\hat{\boldsymbol{\theta}}\right\}. \tag{10}$$

This calculation sets processing capacity to 100 or 300 t/day for every risk row while retaining its other observed predictors. Percentile intervals repeat the standardization for all 1,999 lineage-bootstrap coefficient draws. These are adjusted descriptive probabilities for the observed risk population, not predicted effects of physically enlarging a facility.

The revision-frozen 300-versus-100 contrast is retained for continuity with the model decision record, but its empirical support is explicit. In the broad risk frame, 24, 60, and 120 t/day are the 25th percentile, median, and 75th percentile; 300 t/day is approximately the 99th empirical percentile. Only 315 of 15,154 risk rows and four of 35 events occur at or above 300 t/day. Annual risk is therefore also standardized at 24, 60, and 120 t/day to display the fitted gradient in denser regions of the data.

3.5.4 Sensitivity and design diagnostics

The remaining checks ask whether the scale profile depends on continuity, reporting-state, functional-form, or geographic-concentration choices. They test fragility; they do not convert the association into a causal effect.

The earlier 11-parameter Firth model and conventional links remain specification sensitivities.

Three additional diagnostics retain the same five-parameter structure. The capacity term is replaced by either $\log(1+C)$ or linear $C/100$ and translated to its own 300-versus-100 t/day contrast. The model is also refitted after omitting each of the 21 prefectures containing an event, and on the two-prior-year reporting-state frame. These diagnostics use model-based intervals; they test functional-form, geographic, and state-definition fragility but do not replace the 1,999-lineage-bootstrap primary inference.

Finally, a design audit reports predictor correlations and variance inflation factors (VIFs). Calendar time and logged elapsed risk are correlated because a lineage observed for longer also tends to appear later in the panel. The audit therefore limits interpretation of those temporal coefficients and checks whether that dependence extends to the processing-scale term. The flexible calendar-era and duration-band model remains the corresponding specification sensitivity.

3.6 RQ3: engineering decomposition and component models

3.6.1 Quantities and analysis sample

This section proceeds in two steps. First, it models installed generator size and annual use as separate outcomes. Second, it uses an exact accounting identity to show how those components combine with waste loading to produce gross MWh/t. The identity organizes interpretation; it is not a second source of empirical evidence.

The engineering analysis separates what is installed from how intensively it is used. Let G_{it} be annual gross generation in megawatt-hours (MWh), W_{it} annual throughput in tonnes, K_{it} installed electrical capacity in kilowatts (kW), and C_{it} waste-processing design capacity in tonnes per day (t/day). Define

$$Y_{it} = \frac{G_{it}}{W_{it}} \qquad \text{(gross generation intensity, MWh/t)}, \tag{11}$$

$$D_{it} = \frac{K_{it}}{C_{it}} \qquad \text{(generator design intensity, kW per t/day)}, \tag{12}$$

$$F_{it} = \frac{G_{it}}{8.76K_{it}} \qquad \text{(annual electrical capacity factor)}, \tag{13}$$

$$U_{it} = \frac{W_{it}}{365C_{it}} \qquad \text{(waste-processing utilization)}. \tag{14}$$

The factor 8.76 converts one kW operating for 8,760 hours into annual MWh. These quantities come from annual administrative reports rather than continuous meter or dispatch records. Accordingly, this thesis treats F_{it} as an administrative annual electrical capacity-factor proxy; later references to "capacity factor" use this bounded meaning. The definitions imply the exact facility-year identity

$$Y_{it} = \frac{8.76}{365}\frac{D_{it}F_{it}}{U_{it}} = 0.024\frac{D_{it}F_{it}}{U_{it}}. \tag{15}$$

Gross MWh/t is thus a ratio produced jointly by installed sizing, annual use of that electrical capacity, and annual waste loading. The identity does not say that any one component is exogenous. For example, throughput, capacity factor, maintenance, and waste composition may be jointly determined during a year.

The primary sample requires positive installed capacity, throughput, and gross output. Values are excluded rather than clipped if gross intensity falls outside 0.01–0.80 MWh/t, electrical capacity factor outside 0.02–1.20, waste-processing utilization outside 0.02–1.20, or generator design intensity outside 0.1–100 kW per t/day. A non-missing non-negative reported age and all model fields are also required. Of 6,660 positive-throughput, positive-output rows, 149 fail at least one check, leaving 6,511 rows across 493 lineages. Heating value between 3 and 25 megajoules per kilogram (MJ/kg) is audited as a plausibility field but is not an inclusion condition for the primary component models; 102 otherwise valid rows lack it.

These thresholds are broad plausibility screens, not engineering-performance standards and not cut points selected to optimize the reported estimates. The lower bounds remove zeros and near-zero ratios inconsistent with the intended operating-generator-year frame. The 1.20 upper bounds for capacity factor and waste utilization deliberately allow moderate calendar, reporting, and denominator mismatch instead of treating 1.00 as an error-free physical ceiling. Only 5 of 6,511 retained capacity-factor rows exceed 1.00, and the principal component conclusions remain stable under the conservative-bound sensitivity. The wide design-intensity range permits heterogeneous generator sizing while removing ratios likely driven by unit or field errors. Conservative and broad predefined-bound sensitivities test dependence on these choices; unusual valid observations may still be excluded, and plausible-looking errors may remain.

3.6.2 Primary component models

The first pair of models asks whether reported start-year cohorts differ in raw installed generator size and annual electrical-capacity use after comparing records with the same observed scale, technology, furnace count, and fiscal year. Keeping these outcomes separate prevents gross MWh/t from acting as an ambiguous stand-alone performance score.

Two pooled component models are estimated by ordinary least squares with standard errors clustered by stable lineage (Wooldridge, 2010). Installed electrical capacity is modeled in its raw reported unit:

$$\log K_{it} = \alpha_K + V_{it}'\beta_K + \beta_{KC}\log C_{it} + T_{it}'\eta_K + \lambda_t + \varepsilon_{Kit}, \tag{16}$$

$$\log F_{it} = \alpha_F + V_{it}'\beta_F + \beta_{FC}\log C_{it} + \beta_U U_{it} \\ + T_{it}'\eta_F + \lambda_t + \varepsilon_{Fit}. \tag{17}$$

V_{it} contains reported facility start-year cohorts before 1990, 1990–1999, and 2000–2009 relative to 2010 or later. Reported facility start year is an administrative cohort marker, not a verified boiler or generator installation date. T includes furnace count and coarse furnace and facility-type groups; λ_t are fiscal-year indicators. Because $\log D = \log K - \log C$, the design-intensity scale elasticity is $\beta_{KC} - 1$ under identical controls; this is algebraic, not independent corroboration.

A direct gross-output model replaces $\log C$ with $\log W$ and $\log K$, retaining cohort, technology, furnace-count, and year terms.

3.6.3 Common-control accounting decomposition

This second step is an accounting reconciliation rather than another performance test. By applying exactly the same rows and controls to every component, it allocates each adjusted gross-intensity contrast among installed sizing, annual electrical-capacity use, and waste loading.

Assessing whether sizing makes a substantial contribution requires a method-matched comparison. Four additional component regressions therefore use exactly the same design matrix X_{it}: cohort indicators, log processing capacity, furnace count, coarse furnace and facility types, and fiscal-year indicators. They estimate

$$\begin{aligned} \log D_{it} &= X_{it}'\gamma_D + e_{Dit}, \\ \log F_{it} &= X_{it}'\gamma_F + e_{Fit}, \\ \log U_{it} &= X_{it}'\gamma_U + e_{Uit}, \\ \log Y_{it} &= X_{it}'\gamma_Y + e_{Yit}. \end{aligned} \tag{18}$$

Because $\log Y = \log(8.76/365) + \log D + \log F - \log U$, ordinary least-squares linearity under identical rows and controls implies, for each cohort contrast,

$$\gamma_Y = \gamma_D + \gamma_F - \gamma_U. \tag{19}$$

This common-control identity decomposition is an accounting reconciliation of the relative log-scale components of generator sizing, annual electrical- capacity use, and waste loading. Because the equality is guaranteed by the definitions and identical regression design matrices, it cannot independently confirm the outcome models or establish causal mediation. It is distinct from the primary capacity-factor model above, which includes U and asks how capacity factor differs at equal observed waste utilization. Those conditional capacity-factor coefficients cannot be inserted into the component sum. A sensitivity excludes the 14 lineages whose reported facility start-year cohort changes during follow-up, leaving 6,291 rows across 479 lineages.

3.6.4 Diagnostics, robustness, and exploratory extension

These checks examine whether the component interpretation survives alternative samples, bounds, time periods, and within-episode comparisons. The final pathway comparison remains exploratory because observed administrative labels cannot identify physical project mechanisms.

A specification diagnostic compares a legacy-style regression of $\log Y$ on reported age, processing scale, waste utilization, heating value, technology, and year with the same model after adding $\log D$. Both specifications use the 5,806 engineering-valid rows with reported heating value in the 3–25 MJ/kg plausibility range. This is not a causal mediation design. It asks whether coefficients previously attributed to age, scale, or utilization are stable after the omitted installed-sizing component is represented.

Robustness checks split the period into FY2005–FY2014 and FY2015–FY2024, give each lineage equal total weight, and apply conservative and broad predefined engineering bounds. Asset-episode fixed effects and exact-adjacent first differences are used only for operating components that vary meaningfully within an episode. Design intensity is predominantly a between-asset attribute, so within-episode change is not presented as its primary estimand.

Finally, event pathways are classified from observed administrative continuity. A continuity-lineage entry remains in the same lineage and asset episode across adjacent years without a reported reset. A rebuild/replacement-like entry has an asset-episode, reported-start-year, or mature-to-new age reset. A forward-dated/placeholder entry has a future start year or new-build placeholder name. These labels are descriptive evidence rules. They do not verify the physical project mechanism. First-complete-year outcomes are compared as within-fiscal-year percentile ranks among engineering-valid generators, which reduces confounding by fleet-wide time trends but does not remove selection into each pathway.

4 Results

The results follow the three analytical populations defined in Table 2. In this section, *entry* always means first reported positive installed capacity within an observed administrative lineage, and gross MWh/t remains a composite gross-output ratio rather than an independent efficiency measure.

4.1 RQ1: facility counts understate waste-volume coverage

Answer first. Count-based participation understates the share of observed waste activity covered by generating facilities, while the much smaller change among endpoint-common lineages shows that the all-record trend should not be read as widespread conversion among continuing facilities.

The official 415/991 ratio (41.9%) describes the Ministry's published count of electricity-generating facilities. The results below use the separate analytical definition of positive installed capacity among retained facility records.

Installed-capacity participation rises steadily from 21.6% of facility records in FY2005 to 41.1% in FY2024. Positive-output facilities already handle a much larger share of waste: throughput coverage rises from 60.5% to 80.1% over the same period. The design-capacity share at installed-capacity facilities moves from 56.0% to 70.5%. Figure 1 therefore shows long-run increases with year-to-year variation under all three denominators, alongside a persistent count-volume gap.

The lineage diagnostic changes the interpretation of that 19.50-percentage- point rise. Among 732 administrative lineages observed in both FY2005 and FY2024, installed-capacity prevalence increases only from 29.92% to 32.10%, or 2.19 points. Among 678 endpoint-common lineages retaining the same reported asset episode, it increases from 30.53% to 31.42%, or 0.88 points. The 713 lineages observed in all 20 fiscal years show a similarly modest change, from 29.73% to 32.26%.

Table 3: Endpoint fleet-composition diagnostic

Administrative group	FY	*N*	With capacity	Share
All endpoint records	2005	1,318	285	21.62%
All endpoint records	2024	1,014	417	41.12%
Endpoint-common lineages	2005	732	219	29.92%
Endpoint-common lineages	2024	732	235	32.10%
Endpoint-common, same episode	2005	678	207	30.53%
Endpoint-common, same episode	2024	678	213	31.42%
FY2005-only lineages	2005	586	66	11.26%
FY2024-only lineages	2024	282	182	64.54%

The endpoint-only groups are sharply different: 64.54% of FY2024-only lineages report installed capacity, compared with 11.26% of FY2005-only lineages. Together with the much smaller endpoint-common increase, this pattern is consistent with a substantial contribution from changing observed fleet composition rather than widespread conversion among continuing lineages. It is not an additive decomposition and does not identify physical construction or closure because administrative appearance and disappearance can also reflect reporting and identity history.

The FY2024 cross-section makes the distinction concrete. The administrative panel contains 1,014 facility records, of which 417 report positive installed electrical capacity, giving the 41.1% all-record participation rate. Among the 879 records with positive throughput, 408 report installed capacity, or 46.4%. Holding the installed-capacity state fixed therefore isolates a 5.3-percentage- point denominator difference. Positive output appears in 410 records: 40.4% of all records and 46.6% of positive-throughput records. The two output percentages use the same numerator state and isolate a 6.2-point denominator difference.

The 410 positive-output facilities process 24.70 million of the 30.84 million recorded tonnes, or 80.1%. In contrast, 469 operating non-generators process 6.14 million tonnes, or 19.9%. Another 126 records report neither positive throughput nor generation, and nine installed-capacity records report no positive output. Two FY2024 records report positive output without positive installed capacity. Installed-capacity and positive-output states are therefore reported separately rather than forced into one binary partition.

After the predefined output and capacity-factor checks, valid generator rows cover 79.7% of FY2024 throughput. Their throughput-weighted conditional gross intensity is 0.425 MWh/t. Multiplying 0.797 by 0.425 gives 0.338 MWh per total fleet tonne, exactly matching valid gross generation divided by all recorded throughput. This arithmetic is more informative than the facility count alone: most recorded waste is already processed at facilities reporting positive generation, even though installed equipment appears in fewer than half of facility records.

The result does not imply that the remaining 19.9% of operating throughput can or should all be redirected or equipped. It also does not measure electricity used internally, net export, heat recovery, marginal emissions, or project cost. Its contribution is denominator discipline: installed-capacity participation is 41.1% among all records and 46.4% among active records, while positive-output participation is 40.4% and 46.6%, respectively. Positive- output facilities cover 80.1% of tonnes, and installed-capacity facilities cover 70.5% of design capacity. Matching on activity narrows the count-volume contrast but does not remove it.

4.2 RQ2: entry is rare and associated with processing scale

Answer first. First reported entry remains uncommon across the observed risk set, but its positive association with prior-year processing scale is stable across the reported sensitivity analyses. The age profile is less stable across continuity definitions and does not support an age-only rule.

The descriptive risk set contains 55 first reported installed-capacity entries. The 35 exact-year modeled events comprise 24 continuity-lineage and 11 rebuild/replacement-like entries; none is forward-dated. The four calendar eras contain 7, 4, 14, and 10 modeled events, respectively.

Only 35 events remain after requiring an exact one-year lag and complete prior age and capacity; 33 remain after requiring positive prior-year throughput. This sparse count is central to interpretation. The Firth model addresses finite-estimate and first-order bias problems, but it cannot create information that the panel does not contain.

Observed rates already show strong scale ordering. Entry occurs in 1 of 3,854 risk rows in the smallest processing-capacity quartile (0.026%), 2 of 4,175 in the second quartile (0.048%), 9 of 3,702 in the third (0.243%), and 23 of 3,423 in the largest (0.672%). Because risk duration, calendar period, and age differ across these groups, the adjusted model is needed before interpreting the gradient.

The most defensible translation begins where observations are dense. At the risk-frame 25th percentile, median, and 75th percentile of 24, 60, and 120 t/day, standardized annual predictions are 0.68 (95% confidence interval [CI] 0.35–1.08), 1.37 (0.84–1.97), and 3.29 (2.28–4.57) entries per 1,000 facility-years. The 100 t/day memo comparison level gives 2.53 (1.73–3.52). Entry is uncommon throughout this support-rich range, but the fitted absolute risk rises with processing scale.

The revision-frozen five-parameter broad Firth model gives a coefficient of 2.749 on $\log(1+C/100)$. Its retained 300-versus-100 t/day contrast has an odds ratio of 6.72 with a 1,999-lineage-bootstrap 95% CI of 4.31–12.46. The standardized predictions are 2.53 and 16.66 entries per 1,000 facility-years, a difference of 14.13 (7.38–26.77). The 300 t/day value is an upper-tail prediction at the 98.98th empirical percentile: only 315 rows (2.08%) and 4 modeled events occur at or above it. It is retained as a memo-continuity and tail-sensitivity contrast, not treated as the only or most representative translation of the model.

The prior-operation, same-episode, and identity-certain 300-versus-100 odds ratios are 7.09 (4.08–13.76), 7.15 (4.44–14.05), and 6.76 (4.23–12.30), respectively. Because these frames are nested, their coefficients are parallel sensitivity estimates rather than between-group contrasts.

The broad age coefficient is -0.327 per ten years (bootstrap CI -0.774 to 0.070); prior-operation and identity-certain intervals also span zero. The same-episode estimate is -0.751 (CI -1.364 to -0.206) but uses only 24 events. Age is therefore continuity-sensitive. Reclassifying each event leaves scale odds ratios from 6.12 to 7.30; deleting each event's entire lineage leaves 6.13 to 7.30. No one event or event lineage creates the scale result.

Alternative capacity forms also remain positive. Using $\log(1 + C)$ gives an odds ratio of 5.01 (model-based 95% CI 2.85–8.83), while linear capacity gives 4.22 (2.69–6.61). Omitting each event prefecture gives 6.14–7.18 across 21 fits. The stricter 30-event reporting-state frame gives 6.21 (3.26–11.81). These are diagnostics with model-based intervals; only the revision-frozen primary uses the 1,999-lineage bootstrap for headline uncertainty.

The design audit gives a correlation of 0.909 between calendar time and logged elapsed risk. Their VIFs are 5.76 and 6.15, so those temporal coefficients are not interpreted separately. The processing-scale VIF is only 1.10, and its correlations with calendar time and elapsed risk are 0.013 and 0.039. Temporal dependence therefore qualifies the nuisance time terms but does not explain the estimated scale ordering. The flexible era/duration sensitivity yields a similar 300-versus-100 odds ratio of 6.13.

Table 4: Consolidated entry specification audit

Specification	Events or fits	300-versus-100 t/day odds ratio	Purpose and inference
Revision-frozen five-parameter broad frame	35	6.72 (4.31–12.46)	Primary; 1,999-lineage bootstrap
Prior-operation frame	33	7.09 (4.08–13.76)	Positive prior-year throughput
Same-episode frame	24	7.15 (4.44–14.05)	Episode-boundary continuity
Identity-certain frame	35	6.76 (4.23–12.30)	Uncertain-link exclusion
Flexible era/duration Firth model	35	6.13 (3.92–11.21)	11-parameter temporal-form check; 499 whole-lineage bootstrap replications
Alternative capacity forms	2 fits	4.22–5.01	Functional form; model-based lower CI bounds 2.69–2.85
Two-prior-year reporting state	30	6.21 (3.26–11.81)	Stricter prior-state evidence
Leave-one-event-prefecture fits	21 fits	6.14–7.18	Geographic influence, not confounding control
Event or event-lineage attacks	70 fits	6.12–7.30	Single-event influence

Table 4 distinguishes bootstrap headline inference from model-based and deletion diagnostics. A stable direction across these rows does not recover unobserved technology, finance, policy exposure, or project history omitted from the model.

The supported RQ2 conclusion is narrow and stable across the reported sensitivities. First reported capacity entry is rare and positively associated with processing scale. This concerns witnessed transitions among prior non-generators, not entry across the whole fleet. The administrative data do not establish a general independent age pattern, and scale can proxy for contracts, finances, catchments, technology, geography, or municipal capacity. Neither profile should be used as if the model had identified an equipment-project response.

4.3 RQ3: cohort differences show a substantial installed-sizing component

Answer first. Reported start-year cohorts differ much more clearly in installed generator size than in uniformly lower annual electrical- capacity use. Gross MWh/t therefore cannot distinguish design from operation unless its components are analyzed separately.

The engineering-valid generator frame has 6,511 observations from 493 stable lineages. Median gross intensity is 0.327 MWh/t, median generator design intensity is 14.0 kW per t/day, median electrical capacity factor is 0.607, and median waste-processing utilization

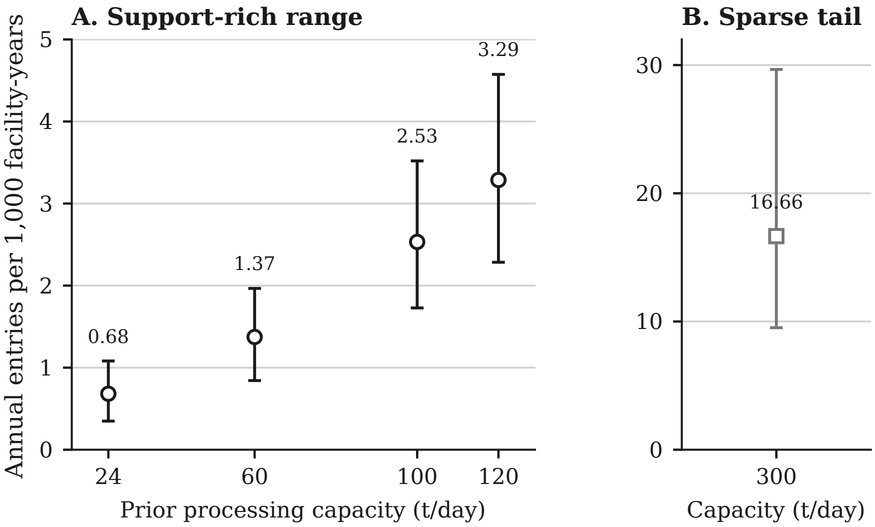

Figure 3: Standardized annual first-entry risk from the five-parameter Firth logistic model, expressed as entries per 1,000 facility-years at specified prior-year processing capacities. Points are model-standardized predictions; vertical bars are 95% intervals from 1,999 whole-lineage bootstrap replications. Panel A shows the support-rich 24–120 t/day range. Panel B uses a different vertical scale to isolate the 300 t/day prediction; 300 t/day is the 98.98th percentile, with only 315 risk rows and four events at or above that level.

is 0.609. These summaries describe generator-years that pass stated bounds, not the full fleet.

The RQ3 conclusion rests first on separate raw installed-kW and annual capacity-factor models. The exact component identity then explains how those independently reported quantities combine; it is not treated as independent evidence that sizing must dominate.

Reported facility start-year cohorts differ much more in installed generator size than in annual capacity factor. Relative to 2010 or later and conditional on observed controls, adjusted installed capacity is 79.1% lower before 1990, 58.6% lower in 1990–1999, and 23.5% lower in 2000–2009.

The installed-kW elasticity with respect to processing t/day is 1.532 (95% CI 1.447–1.617), and the coefficient of determination (R^2) is 0.786. Subtracting one gives the 0.532 design-intensity elasticity under identical controls; this is algebraic rather than independent replication.

The electrical-capacity-factor model tells a different story. Conditional on processing scale, waste utilization, technology, furnace count, and fiscal year, pre-1990 and 1990s capacity factors are 35.3% (95% CI 24.6–46.8%) and 22.0% (14.4–30.0%) higher. The 2000s estimate is 1.5% higher, with a CI from 4.1% lower to 7.4% higher. Waste utilization is positively associated with log electrical capacity factor (coefficient 1.695, 95% CI 1.448 to 1.942), while processing scale is negatively associated (-0.116, 95% CI -0.162 to -0.070). This model explains 33.9% of log capacity-factor variation. These coefficients describe annual use of installed kW. They do not show that utilization has an independent positive association with gross MWh/t after generator sizing is represented.

The common-control decomposition provides a narrower accounting reconciliation. Every row in Table 5 uses the same 6,511 observations and the same cohort, processing-scale, technology, furnace-count, and year controls. The three component columns are log differences from the 2010-or-later cohort and sum exactly to the directly fitted log gross-intensity difference.

Table 5: Common-control cohort decomposition of log gross MWh/t

Cohort	Design γ_D	Capacity factor γ_F	Waste loading $-\gamma_U$	Total γ_Y
Before 1990	-1.565	+0.016	+0.299	-1.250
1990–1999	-0.883	+0.020	+0.172	-0.690
2000–2009	-0.267	-0.094	+0.089	-0.272

Installed sizing is the largest absolute point-estimate accounting component in every older cohort. For the pre-1990 and 1990s cohorts, common-control capacity-factor differences are near zero; their positive capacity-factor coefficients in the primary model arise after conditioning on waste utilization and answer a different equal- utilization question. For the 2000s cohort, both smaller sizing and a lower common-control capacity factor contribute. Excluding all 14 cohort-switching lineages leaves 6,291 rows across 479 lineages and produces the same ordering, with direct log gaps of -1.306, -0.702, and -0.280. These exact conditional sums are accounting attribution, not causal mediation.

The direct gross-output model is consistent with the component structure. The elasticity of annual gross MWh with respect to throughput is 0.638 (95% CI 0.536 to 0.740), and the elasticity with respect to installed electrical capacity is 0.576 (95% CI 0.502 to 0.650), conditional on cohort, observed technology, furnace count, and year. The model R^2 is 0.914. It shows that both waste loading and installed kW are associated with annual output, but it is not a production function with exogenous inputs.

An accounting-consistency diagnostic compares gross-intensity specifications on the 5,806-row plausible-heating-value subset. In the legacy-style model without generator design intensity, the reported-age coefficient is -0.0349, the processing-capacity coefficient is 0.1001, and the waste-utilization coefficient is 0.6699; all have $p < 0.001$. After adding log generator design intensity, the corresponding coefficients become -0.0020 ($p = 0.2977$), -0.0092 ($p = 0.1991$), and -0.0995 ($p = 0.2038$). The sizing coefficient is 0.7532 ($p < 0.001$), and model R^2 rises from 0.4737 to 0.8131. This is not a causal decomposition, and the increase is partly expected because design intensity is algebraically related to gross MWh/t. The diagnostic shows that the former age, scale, and utilization interpretation is sensitive to whether installed sizing is represented; it does not independently prove that sizing causes the cohort pattern.

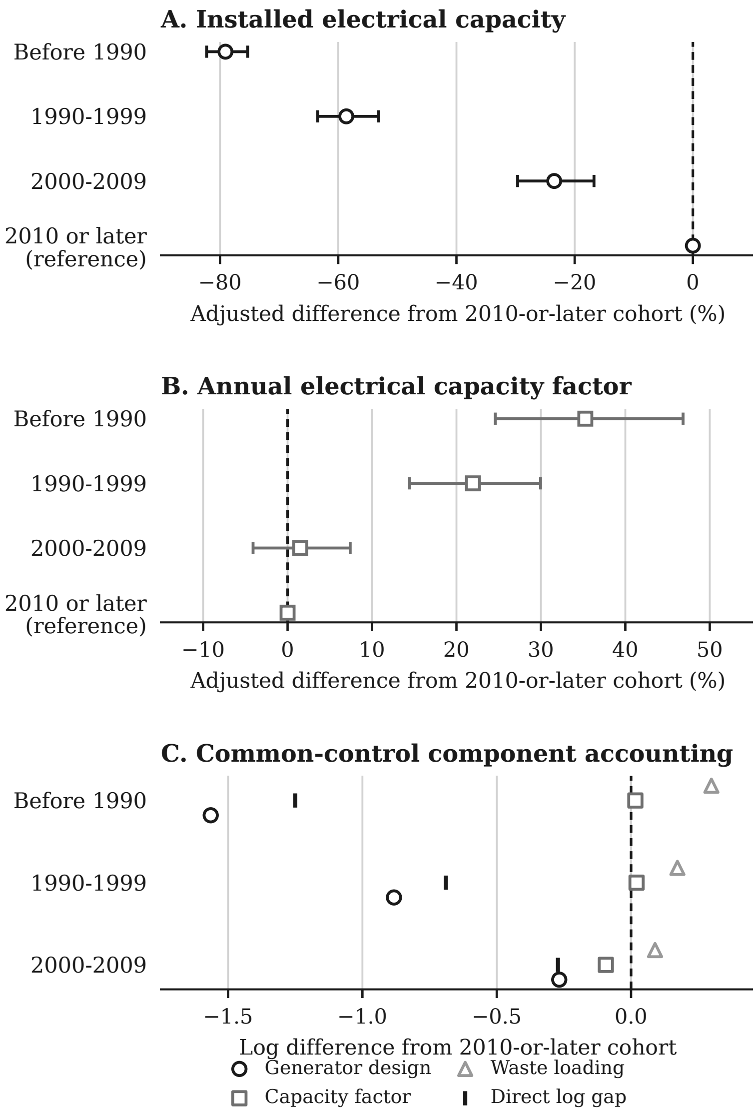

Figure 4: Reported facility start-year cohort evidence for 6,511 engineering-valid facility-years from 493 administrative lineages. Panels A and B show adjusted percentage differences from the 2010-or-later cohort in installed electrical capacity and annual electrical capacity factor; bars are lineage-clustered 95% confidence intervals. Panel C uses the same sample and common controls to reconcile the direct log gross-MWh/t gap into generator design, capacity-factor, and waste-loading components. Those components sum by an exact accounting identity and do not estimate causal mediation. Reported facility start year is not a verified generator-installation date.

Adjacent-year rank persistence is highest for generator design intensity: 0.995 across 5,963 pairs from 470 lineages. Gross-intensity rank persistence is 0.961 and capacity-factor persistence is 0.873. The corresponding within-to-total variance ratios are 0.016 for log design intensity, 0.089 for log gross intensity, and 0.426 for log capacity factor. Sizing is therefore mostly a between-asset design attribute, while annual capacity factor contains much more within-asset movement.

The principal component conclusions remain stable across the two ten-year windows, lineage-equal weighting, and conservative or broad engineering bounds. For example, the processing-scale coefficient in the design-intensity model is 0.474 in FY2005–FY2014 and 0.577 in FY2015–FY2024; it is 0.520 under conservative bounds and 0.536 under broad bounds. Excluding every identity-uncertain lineage leaves 6,450 rows across 487 lineages and gives a nearly unchanged coefficient of 0.533. Within-asset-episode models also retain a positive association between utilization and electrical capacity factor. Those checks strengthen the component description but do not resolve simultaneous changes in throughput, maintenance, installed capacity, and output.

5 Discussion

5.1 What the thesis changes in the fleet narrative

Each analytical margin corrects a different intuitive but incomplete reading of the fleet. Table 6 separates the tempting interpretation from the strongest statement supported by the evidence.

Table 6: Interpretive corrections produced by the three-margin design

Common reading	Evidence-based correction	Defensible interpretation
Only about 41% generate, so recovery reaches little waste.	Positive-output facilities handle 80.1% of throughput; installed-capacity facilities represent 70.5% of design capacity.	Record prevalence is incomplete, but generation is concentrated in larger or more active parts of the observed fleet.
The 19.50-point rise means widespread retrofit of incumbents.	The rise is 2.19 points among 732 lineages observed at both endpoints.	Changing observed composition is substantial, but the comparison is not an additive decomposition and does not verify retrofit projects.
Newer cohorts are simply more efficient.	Older cohorts have much smaller adjusted installed kW, but not uniformly lower capacity factors; sizing is the largest absolute point-estimate component.	Gross MWh/t combines sizing, annual capacity use, and waste loading; the component identity is descriptive, not causal.

The transition model adds a separate correction. Processing scale is positively associated with first reported entry in the primary model and all sensitivity frames, but the model does not show that enlarging a facility would cause entry. Scale can represent unobserved municipal, technical, and financial conditions. Age is continuity-sensitive and therefore cannot support a simple age-only rule.

5.2 Direct answers to the research questions

RQ1 is answered by matched state and denominator contrasts, not by one preferred percentage. In FY2024, positive-output facilities handle 80.1% of recorded throughput and installed-capacity facilities hold 70.5% of processing design capacity, despite installed capacity appearing in only 41.1% of all records. Generation is therefore concentrated in larger or more heavily used parts of the observed fleet. The contrast does not mean that 80.1% of waste becomes electricity or that the covered facilities are environmentally optimal. It also separates prevalence from incumbent diffusion: the all-record endpoint increase is 19.50 percentage points, compared with 2.19 points among 732 lineages observed in both endpoint years.

RQ2 is answered by a scale association stable across the reported sensitivity analyses and a continuity-sensitive age association. In the support-rich 24–120 t/day range, standardized annual predictions rise from 0.68 to 3.29 entries per 1,000 facility-years. The revision-frozen 300-versus-100 t/day odds ratio is 6.72, and Table 4 shows that the positive scale gradient survives the continuity, functional-form, reporting-state, geographic, and event-influence checks. The 300-t/day prediction is nevertheless a tail contrast, supported by 315 risk rows and four events at or above that level. Age is less stable across continuity definitions and should not support an age-only screening rule. The outcome remains first reported positive installed capacity within an administrative lineage, not a verified retrofit, construction start, or causal response to changing capacity.

RQ3 is answered by separating installed design from annual use. Older reported start-year cohorts have much smaller adjusted installed electrical capacity, but not uniformly lower administrative capacity-factor proxies. Gross MWh/t therefore combines generator sizing, annual capacity use, and waste loading rather than acting as an independent operating-efficiency score. Under shared controls, sizing is the largest absolute point-estimate accounting component of every older-cohort gap. The exact component sum is reconciliation by construction, not independent evidence, and the small pathway comparison remains a hypothesis-generating appendix rather than an answer to RQ3.

Together, the answers provide one ordered diagnosis. RQ1 locates electricity recovery within the fleet; RQ2 describes which observed non-generator lineages first enter the installed-capacity state; and RQ3 explains the observable structure inside the generating segment. No result is asked to answer a question belonging to another frame. That separation is the central thesis contribution and the main safeguard against an aggregate or conditional result being interpreted causally.

5.3 Evidence-bound implications

Monitoring should report facility, throughput, and design-capacity coverage together. Screening can treat processing scale as a marker for further feasibility work, but not use an age-only rule. Observed generator-years that meet the engineering-validity criteria should be compared conditional on sizing and reported facility start-year cohort before gross MWh/t is interpreted as an operating gap.

These are measurement and diagnostic implications, not a ranked list of projects. The study does not determine whether a municipality should build, replace, coordinate, maintain, or retire a facility. Such decisions require capital costs, waste-supply agreements, grid and internal-use conditions, maintenance history, emissions controls, heat demand, and alternatives higher in the waste hierarchy. The thesis's role is to prevent an aggregate statistic or misspecified ratio from deciding those questions implicitly.

5.4 Limitations and next evidence needed

The 20 workbooks are hash-identified and parser mappings are documented, but their original retrieval timestamps and publisher-side revision history are unavailable. Future acquisition should archive both.

Stable administrative lineages remain inferred. The resolver addresses exact duplicates, code breaks, row-order dependence, and known difficult links, but no physical-site registry verifies ownership, construction, or closure histories. The blinded clerical-review packet is generated, but an independent reviewer has not yet completed and adjudicated it. This outstanding human validation gate is why the thesis describes deterministic linkage and sensitivity checks rather than independently audited identity. Administrative absence is therefore not modeled as a physical outcome.

Entry is rare and partially left-censored. Firth estimation and lineage bootstrapping cannot overcome only 35 broad-frame, 33 prior-operation, and 24 same-episode events. Although all 1,999 requested bootstrap replications per frame converge, resampling cannot create missing project information. The parsimonious models omit financing, prices, contracts, catchments, maintenance, and detailed technology. Scale may proxy for these, so odds ratios are associations rather than effects of changing t/day.

Installed-capacity state is also a reporting construct. Forty-nine panel rows across six lineages report positive gross output despite a blank or zero installed-capacity field. None is the immediately prior row of a modeled event, and the two-prior-year sensitivity remains positive, but these checks do not prove that blanks represent physical absence. External equipment or commissioning records are needed to verify event meaning.

Generator fields also have strict boundaries. Gross generation is not net export; on-site use and useful heat are incomplete; reported capacity may differ from availability;

and MWh/t is not the European Union R1 energy-efficiency indicator, a lifecycle measure, or an economic measure (Grosso et al., 2010). Heating-value coverage is insufficient for a common thermal measure. Bounds remove 149 rows but may exclude unusual valid cases or retain plausible-looking errors.

Reported facility start year is not an equipment date and can bundle original design, later equipment, reporting, waste, and municipality context. Annual throughput, capacity factor, maintenance, and output are jointly determined. Controls, fixed effects, and adjacent differences do not create exogenous variation; the equations remain accounting identities and conditional descriptions.

The 2010-or-later cohort enters only from FY2010 onward. Fiscal-year indicators create within-year comparisons where cohorts overlap, but cannot remove selective survival or unrecorded replacement.

The next evidence programme has three priorities. First, commissioning, procurement, or equipment records should verify the 35 modeled events and the administrative lineages around them. Second, net-export, internal-use, useful- heat, outage, cost, and waste-composition records should characterize project and operating performance. Third, only after policy exposure and credible comparison units are observed should a later study attempt causal evaluation of FIT, subsidy, retrofit, or replacement effects.

Finally, the pathway contrast is small and selected, and reported resets do not verify physical replacement. It remains a hypothesis-generating pointer toward the first and second evidence priorities, not a project-effect estimate.

6 Conclusion

Japan's fleet looks different at three margins. In FY2024, installed capacity appears in 41.1% of all records and 46.4% of positive-throughput records; positive output appears in 40.4% and 46.6%, respectively. Positive-output facilities handle 80.1% of throughput, and installed-capacity facilities represent 70.5% of processing design capacity. The 19.50-point all-record rise since FY2005 is only 2.19 points among endpoint-common lineages, a contrast consistent with a substantial fleet-composition contribution. At the transition margin, standardized annual predictions rise from 0.68 to 3.29 entries per 1,000 facility-years across the support-rich 24–120 t/day range. The revision-frozen 300-versus-100 t/day odds ratio is 6.72, but 300 t/day is a thinly supported tail contrast. The scale direction is stable across the reported sensitivity analyses, but age remains continuity-sensitive and neither association is causal. At the component margin, independent raw-kW and capacity-factor models show that cohort differences are much larger for installed sizing than for uniformly lower annual use. A shared-control identity then reconciles sizing as the largest absolute point-estimate accounting component of each older-cohort gross-intensity gap; it does not independently confirm a causal mechanism.

The thesis's contribution is therefore not a claim that every non-generator is an equal opportunity or that newer facilities simply perform better. It is a three-margin diagnostic: reconstruct lineages before transitions, separate counts from covered activity, and separate installed design from annual use. That architecture gives a professor or later reviewer a clear foundation for deciding which next question requires engineering, capital-history, or causal evidence.

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The author thanks Prof. Han Ji for supervision and critical feedback during the development of this thesis.

Research Transparency

Data and ethics

This study uses publicly released administrative facility data and does not involve human participants, animal subjects, or private personal data.

Data availability

The facility-level source data are derived from the Ministry of the Environment Japan General Waste Treatment Survey, which is publicly released by the Ministry and e-Stat. Code, manifests, derived tables, figures, and the source workbooks used here are available at `https://github.com/Pann13223029/incineration-paper`. e-Stat permits reuse and modification with source citation under terms compatible with Creative Commons Attribution 4.0.

The thesis itself contains the data definitions, sample construction, model equations, variable definitions, principal estimates, robustness summaries, and interpretive limitations needed to assess its argument. The repository is an optional reproducibility archive and is not required to understand the thesis.

Use of generative artificial intelligence (AI) and AI-assisted technologies

During the preparation of this thesis, the author used OpenAI Codex and Anthropic Claude for language revision, thesis organization, and assistance with code development and review. The author executed the analyses, inspected the source data and generated outputs, independently checked the reported results, reviewed and edited all AI-assisted material, and takes full responsibility for the content. These tools were not used as authors and did not replace author judgment or accountability.

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A Exploratory post-entry pathway comparison

This secondary analysis is outside the three research questions because the pathway labels do not verify physical project types and the followed groups are small. It is retained for hypothesis generation and professor discussion, not as a main thesis finding. The first-complete-year comparison asks where entrants sit in the same-year generator distribution, not what entry causes. Across the 44 exact-year entrants with an engineering-valid outcome one year after entry, mean within-year ranks are 51.6% for gross intensity, 48.1% for generator design intensity, and 56.3% for capacity factor. The pooled entrant therefore lies near the middle of the contemporaneous generator distribution.

The administrative pathways are heterogeneous. One year after entry, 27 continuity-lineage entrants average 0.260 MWh/t and rank at 40.2% for gross intensity, 36.8% for generator design intensity, and 53.8% for capacity factor. Eleven rebuild/replacement-like entrants average 0.442 MWh/t and rank at 72.5%, 66.1%, and 65.5%, respectively. Six forward-dated or placeholder observations are omitted from Figure A.1 because that category is too sparse and its timing is difficult to interpret.

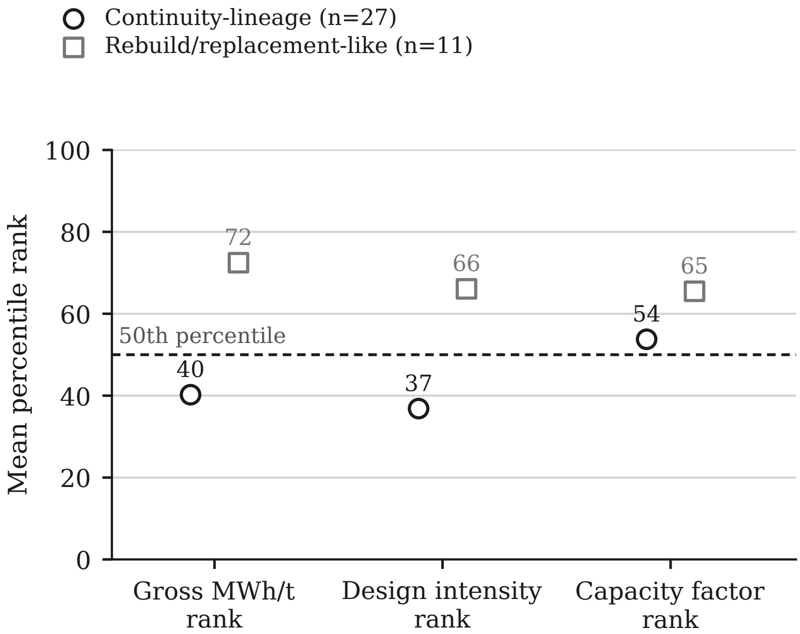

Figure A.1: Mean within-fiscal-year percentile ranks in the first complete year after reported entry for continuity-lineage entrants ($n = 27$) and rebuild/replacement-like entrants ($n = 11$). Metrics are gross MWh/t, generator design intensity (installed kW per t/day of processing capacity), and annual electrical capacity factor among engineering-valid generator-years; the dashed line marks the 50th percentile. The forward-dated/placeholder category ($n = 6$) is omitted because support is sparse. All contrasts are descriptive.

The larger descriptive pathway difference aligns more closely with generator sizing than with capacity factor. This is not an estimated pathway effect: assignment relies on reported resets, follow-up is selective, and there is no counterfactual comparison between otherwise equivalent projects.

B Model coding and focal estimates

B.1 First-entry model

The first-entry likelihood includes only rows with $H_{it} = 1$: at the start of year t, lineage i has no earlier observed positive installed-capacity record. The four reported covariates are age per ten years, $\log(1 + C/100)$ for prior-year processing capacity C in t/day, calendar time per five years centred at FY2014.5, and $\log(1 + R)$ for elapsed observed years at risk. Table B.1 reports every non-intercept coefficient from the primary five-parameter Firth model and its three sensitivity frames. Standard errors are fitted-model estimates; confidence intervals are percentiles from 1,999 whole-lineage bootstrap replications.

The coefficient scale matters. The age estimate is a change in log odds for a ten-year increase in prior reported age. Its broad-frame point estimate of -0.327 corresponds to roughly 28% lower odds, but the bootstrap interval includes no association; it should not be converted into an age rule. The processing-capacity coefficient applies to the transformed variable $\log(1 + C/100)$, not to a simple 100-tonne-per-day increment. The 300-versus-100 t/day odds ratio is therefore calculated as $\exp(2.749 \log 2) = 6.72$. Standardized risks are easier to interpret because they translate the same fit back to annual entries per 1,000 facility-years.

The two uncertainty summaries answer related but different questions. Model-based standard errors describe the curvature of the fitted penalized likelihood. The bootstrap intervals repeat the complete estimation after resampling whole lineages, preserving within-lineage dependence and showing how the estimate varies when the observed lineage composition changes. The thesis uses the 1,999-lineage bootstrap for headline inference and retains model-based quantities for transparent comparison and diagnostics.

Calendar time and elapsed observed risk remain in the model to prevent the capacity coefficient from absorbing a simple panel-time pattern. They are not interpreted separately because their strong correlation makes that separation unstable. Technology and geography are omitted from the primary model because 35 events cannot support a large adjustment set; the leave-one-prefecture and other sensitivity fits assess concentration but do not repair unmeasured confounding.

Table B.1: Firth first-entry model focal coefficients

Frame	Term	Estimate	Model SE	Bootstrap 95% CI	Events
Broad exact-year (primary; 15,154 rows; 1,137 lineages)	Age per 10 years	-0.327	0.214	-0.774 to 0.070	35
Broad exact-year	Log processing capacity	2.749	0.429	2.108 to 3.639	35
Broad exact-year	Calendar time per 5 years	0.696	0.277	-0.012 to 1.149	35
Broad exact-year	Log elapsed risk	-0.638	0.579	-1.610 to 1.116	35
Prior operation (13,072 rows; 1,019 lineages)	Age per 10 years	-0.323	0.231	-0.793 to 0.147	33
Prior operation	Log processing capacity	2.825	0.487	2.030 to 3.783	33
Prior operation	Calendar time per 5 years	0.652	0.318	-0.097 to 1.055	33
Prior operation	Log elapsed risk	-0.497	0.643	-1.469 to 1.347	33
Same asset episode (15,095 rows; 1,135 lineages)	Age per 10 years	-0.751	0.271	-1.364 to -0.206	24
Same asset episode	Log processing capacity	2.838	0.538	2.150 to 3.813	24
Same asset episode	Calendar time per 5 years	0.463	0.346	-0.544 to 0.916	24
Same asset episode	Log elapsed risk	0.047	0.740	-1.190 to 2.663	24
Identity certain (15,107 rows; 1,130 lineages)	Age per 10 years	-0.328	0.214	-0.791 to 0.065	35
Identity certain	Log processing capacity	2.758	0.429	2.081 to 3.620	35
Identity certain	Calendar time per 5 years	0.694	0.277	-0.017 to 1.146	35
Identity certain	Log elapsed risk	-0.639	0.580	-1.552 to 1.067	35

The intercept is included in every fit but is not substantively interpreted. Calendar time and elapsed risk are retained as nuisance controls and are not interpreted separately because they are strongly correlated. Only the broad exact-year frame is the primary model.

B.2 Engineering outcome models

The engineering models use 6,511 generator-years from 493 lineages and lineage-clustered standard errors. Cohort contrasts use 2010 or later as the reference. Shared controls are log processing capacity, furnace count, furnace group, facility group, and fiscal-year indicators. The omitted categories are fluidized-bed furnace, gasification/melting facility, and FY2005; these are coding references rather than normative comparison standards. The capacity-factor model additionally includes waste-processing utilization. The gross-output model replaces log processing capacity with log annual throughput and log installed capacity.

Table B.2: Engineering model focal coefficients

Outcome	Term	Estimate	Clustered SE	95% CI	R^2
Log installed capacity (kW)	Before 1990 cohort	-1.565	0.084	-1.730 to -1.399	0.786
Log installed capacity (kW)	1990–1999 cohort	-0.883	0.063	-1.007 to -0.758	0.786
Log installed capacity (kW)	2000–2009 cohort	-0.267	0.043	-0.352 to -0.183	0.786
Log installed capacity (kW)	Log processing capacity	1.532	0.044	1.447 to 1.617	0.786
Log capacity factor	Before 1990 cohort	0.302	0.042	0.220 to 0.384	0.339
Log capacity factor	1990–1999 cohort	0.198	0.032	0.135 to 0.262	0.339
Log capacity factor	2000–2009 cohort	0.015	0.029	-0.042 to 0.072	0.339
Log capacity factor	Log processing capacity	-0.116	0.024	-0.162 to -0.070	0.339
Log capacity factor	Waste-processing utilization	1.695	0.126	1.448 to 1.942	0.339
Log gross generation (MWh)	Log annual throughput	0.638	0.052	0.536 to 0.740	0.914
Log gross generation (MWh)	Log installed capacity	0.576	0.038	0.502 to 0.650	0.914

Table B.2 reports every focal coefficient used for the RQ3 interpretation. Furnace-group, facility-group, furnace-count, fiscal-year, and intercept coefficients are nuisance adjustments rather than thesis estimands. Sections 3.5.1–3.5.3 define the models and variables; Tables 4–5 and Appendix Tables B1–B2 report the estimates needed to evaluate the three principal research questions without consulting external files.